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# Comparative Analysis of Reconfigurable Low Pass Filter using Biquad Active, Ladder $G_m$-C and Multiple Loop Feedback Techniques

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*Abstract*—**An important concern for LPF with switchable attributes at baseband is to select a appropriate filter structure. Among the different approaches three approaches are prominently used named LC ladder simulation biquad cascade and multiple loop feedback (MLF). This study aimed at comparative analysis of three different low pass filter approaches. The filters are realized using Biquad Technique, ladder technique and MLF approach. Simulations are done using different CMOS technology. In this paper a comparison is made using different parameters. In general, the results prove that a good agreement between all the three approaches usng SPICE simulations.**

***Keywords: Biquad filter, MLF, LC- ladder, MOS-C and reconfigurable low pass filter***

## I. Introduction

The market of smart gadgets has been growing internationally. The interest of multi-standardcompatibility on portable/remote receivers is developing moreand more important these days. In present era wireless system plays a crucial role in our daily life and critical role will be played by Analog baseband (ABB) circuits, with low pass filters (LPFs) and amplifiers with programmable gain [1-3].

Analog base band circuits are required for rejecting the unwanted signal also for increase the amplitude and power for the wanted signals with minimun distortion. The second requirement is to reject noise signals. To cover all standards a filter should have configurable attributes.

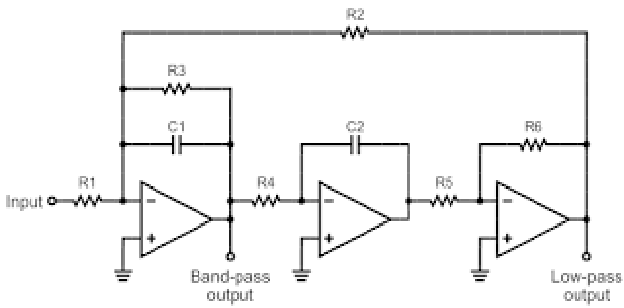
*Analog Filters are Categorized Into:* Active RC filters, MOS-C and filters with OTA-C. Simulations of higher order filter circuits with OTA-C may be done using three different methods. The first method is the cascade of biquadratic sections which can be used to overcome low levels of power and frequencies. Basically these filters use operational amplifier as one of its integral part. The second method is based on ladder prototypes using LC circuit. Ladder circuit generally designed merely on the basis of series and parallel connections of various components and devices. The third method is multiple loop feedback. The biquadratic sections can realize any arbitrary transmission zeros but has a very high sensitivity to components variations.The LC ladder has a verylow sensitivity.Op amp as an integrator is used in multiple feedback filter.

## II. Biquad

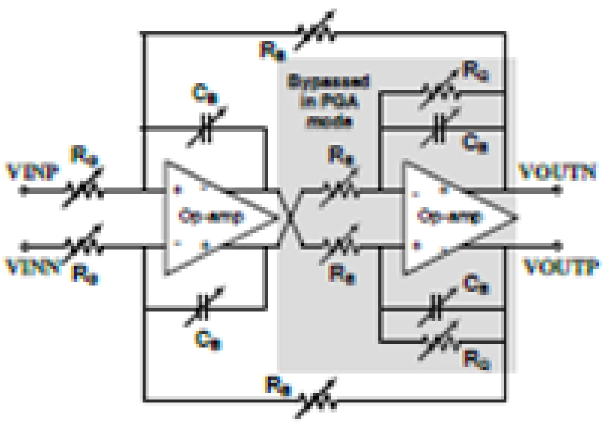
To overcome low levels of power and frequencies active filter topology used which configure band pass filter is as shown in fig 1 (a) and (b), fig 1(a) shows three opamp based Tow-Thomas biquad, fig 1 (b) Tow-Thomas includes two Operational amplifier with one resistor which is connected in feedback path and bank of capacitor (R banks and C banks).The biquad filter consists independent second order sections which offers complex conjugate poles due to this filter response highly dependent on the variation of component values.

The gain of biquad filter will depend on the factor decided by $R_B$=$R_G$. The quality factor will be decided by $R_Q$=$R_B$. With the help of R banks three modes of operational amplifier current can be established. Close-loop BW should not be changed for that the capacitor banks $C_B$ are changed with $R_B$. In fig.1 close-loop BW requirement meets by the gain bandwidth of operational amplifier. In the core Opamp a large GBW required for a good band width in active RC topology [4], where G is gain, Q is quality factor, and fc is BW of the biquad respectively.

The value of the gain bandwidth product of the main Opamp depends on cut off frequency. For a smaller cut off frequency GBW can be relaxed, the benefit of reducing GBW is that power is also reduced in comparison of higher cutoff frequency circuits [4-6]. Similarly, if a filter is designed for low gain value and smaller order the Opamp current could be also reduced resulting smaller quality factor.




(a)




(b)

Fig. 1: (a) Three Opamp Tow-Thomas Biquad [2] (b) Reconfigurable Tow-Thomas Biquad Topology [2]

## III. Ladder

Figure 2 shows the block diagram of elliptical low pass filter which is designed on the leap-frog Gm–C RLC ladder topologyIt is highly tunable. Switched-capacitor array is used as load capacitor of each OTA. In passive ladder approach inductors are used. In LC ladder approach active circuits are implemented for replacing inductors. Passive prototypes offers high sensitivity which is not required. Although some filter responses can not be implemented by using active circuits.

One of the important parameter is the topology architecture on which the performance of the LPF depends. Cut off frequency, filter order and transition band ratio, as well as power consumption these are other performance parameters of this LPF, which can also be reconfigured in this topology which is difficult in traditional cascaded biquad implementation. Flexibility is also required in designing of switchable order filter, but LC ladder approach does not offer high flexibility. The performance of low pass filter can be changed according to three applications WLAN, WCDMA, and Bluetooth, respectively [7, 8].

In all the three applications cut-off frequencies, transition band ratios and power consumptions are different. In this circuit there are three nodes one for each application. Process variation is less in this topology which is very advantageous in designing of low pass filter.

## IV. Multiple Loop Feedback

In early 90s research has been done of using MLF to some extent. There are some flaws which can not be rectified in multiple loop feedback based approach due to that flaws this approach was not very popular.In early 90s switchable order filter design based on circuit approach but at present time it is based on system approach. System approach for switchable order low pass filters is very popular because it offers flexibility and also offers possibility of reusing the existing circuits. Performing synthesis of MLF approach is more complicated in comparison of LC ladder based simulation and biquad approach. Previously the results of MLF approach were only verified through simulations. At present time simulations can be tested with the help of fabrication process. The block diagram of an nth-order LPF is shown in figure 3.In MLF approach design can be implemented by using less number of components in comparison of other approaches, also it offers low sensitivity. In MLF switches are used one is ground switch and the other is series switch but when order is increased non ideal ties of switches will come into the picture.

In the given circuit MLF [9, 10] topology with inverse follow the leader feedback (IFLF) structure is used. In this circuit a classical single stage differential amplifier is implemented with cross coupled loading technique. The advantage of using this cross-coupled load is that it will show the characteristic of negative resistance.

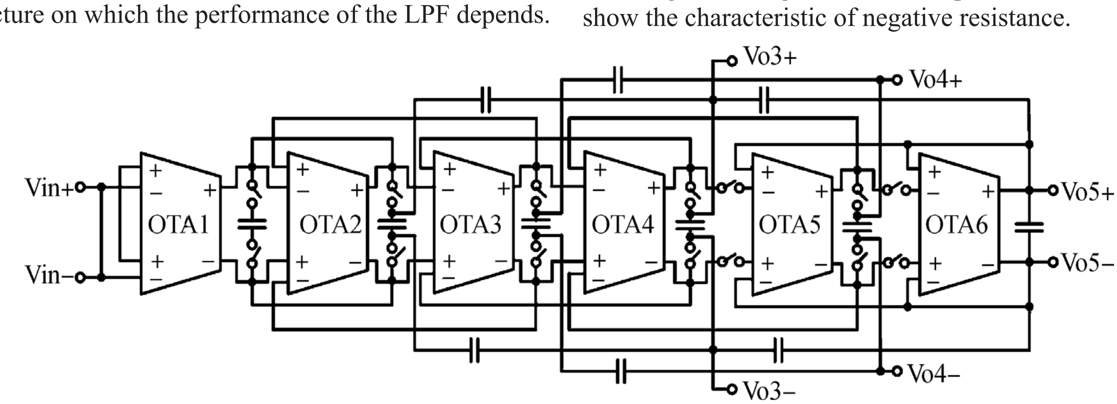



Fig. 2: Block Diagram of Ladder based Low Pass Filter

Table 1: Comparative Analysis of Various Topology

| Ref | Topology | Type | Technology | VDD (Volts) | Power (mW) | IRN (nV/sqrt Hz) | Order | BW (MHz) | Chip area (mm2) |
|---|---|---|---|---|---|---|---|---|---|
| [1] | Biquad Active RC | Chebyshev | 0.13µm | 1 | 3-7.5 | 85.52 | 1/3/5 | 1-20 | 1.94 |
| [2] | Biquad Active RC | Tow Thomas | 65nm | 1.2 | 1.72-9.6 | 87.24 | 2/4/6/8 | 0.2-20 | 0.8 |
| [3] | MLF | Chebyshev | 65nm | 1.2 | 19.8 | 5 | 5/7 | 50 | 0.18 |
| [4] | Ladder Gm-C | Butterworth & Elliptical | 0.35 µm | 2 | 3.8 | 89.42 | 3 | 42 to 215 | 0.074 |
| [18] | Ladder Gm-C | Butterworth | 0.13 µm | 1.2 | 3.3 | 40 | 3/5 | 0.1 to 2.75 | 0.57 |
| [19] | MLF | Elliptical | 0.18 µm | 1.8 | 63 | 0.27 | 5 | 30 | 0.061 |
| [20] | MLF | Chebyshev | 65nm | 2.2 | 20.1 | 6 | 5/7 | 40 | 0.2 |

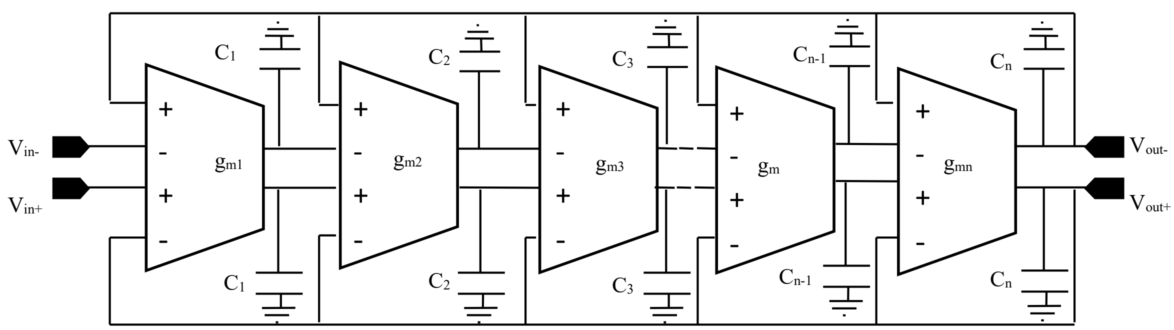



Fig. 3: MLF IFLF LPF All Pole Structure

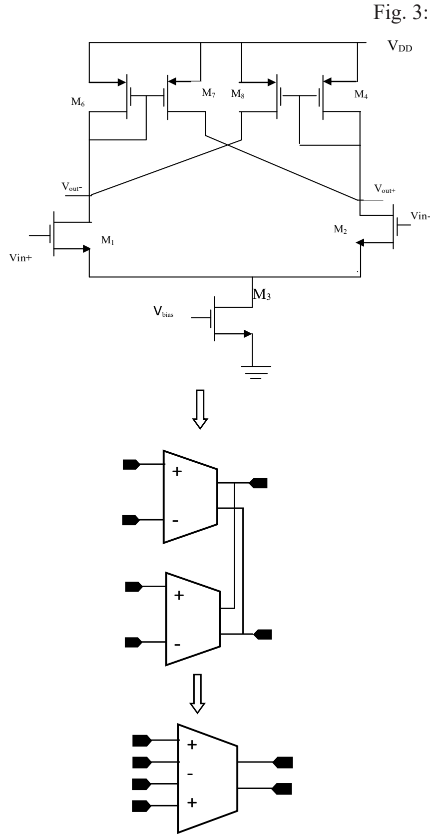



Fig. 4: $g_m$ Cell Circuit of MLF Approach

The open loop DC gain of the $g_m$ cell can be varied and depends on the device size of $M_6$ and $M_7$ as shown in figure 4 It can be significantly boosted by the proper selection of device size. The outputs of 2 identical input cell are connected in parallel to implement a -input cell.

## V. Comparison between System Approaches

Selection an appropriate system approach, for switchable-attributes LPF is an important issue.

Among the various approaches, three are most popular and very much used for research, the LC-ladder simulation, biquad cascade [11] and multiple-loop feedback (MLF).

The LC-ladder simulation method in which inductors have been replaced with active circuits.All the frequency responses can be simulated.But there is also a drawback of replacing inductors that is low sensitivity.

However, there are some limitations using this approach in comparison of ladder filter approach. By using MLF approach some filter functions (e.g. linear-phase responses) cannot be realizable.

Another issue with this approachis flexibility it is least flexible than the passive prototype. For desining this approach several filters of different orders are required. Cascading of biquad sections of second order is not a easy task. For designing reconfigurable order filter this approach provides a simple structure. But there is also a drawback with this approach that the filter response depends on the components value [11].

Multiple loop feedback approach is the most popular approach in active filter design. For implementing this approach $g_m$ cells are cascaded and different feedback paths are provided between grounded capacitors and cells. For designing of reconfigurable order low pass filter the design of each $g_m$ cell should be identical. By using this approach a switchable order filter can be implemented. For changing the order simply the corresponding $g_m$ cells, capacitors and feedback loops should be ON and OFF. Due to this advantageous feature, MLF-based approaches are preferred for implementing reconfigurable order low pass filters.

## VI. Conclusion

In this paper, three different approaches of low pass filter have been compared. All of them are having reconfigurable features. Parameters analyzed are bandwidth, filter-order, power and IRN. With reconfiguration feature maximum performance flexibility can be achieved. Based on the different B.W. and power requirements of each topology chip area can be remarkably optimized.

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